Assignment – 1.19

Switch and Router Configuration Basics Theory

Q - 1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

- **Create VLANs:**
  - In Cisco Packet Tracer, add a switch and enter these commands:

    enable

    configure terminal

    vlan 10

    name Sales

    exit

    vlan 20

    name Support

    exit

<u>Q - 2.</u> Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint: Use the interface range command to select multiple ports efficiently.

➢ <u>Assign ports to VLANs:</u>

interface range fa0/1-3

switchport mode access

switchport access vlan 10

exit

interface range fa0/4-5

switchport mode access

switchport access vlan 20

exit

- **Verify VLANs:**

  show vlan brief

- VLAN 10: Sales — Ports 1, 2, 3
- VLAN 20: Support — Ports 4, 5

Q - 3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

- **Configure static routing:**

  Connect two routers and enter the following commands.

  - **Router 1:**

    enable

    configure terminal

    interface g0/0

    ip address 10.0.0.1 255.255.255.252

    no shutdown

    exit

    ip route 192.168.2.0 255.255.255.0 10.0.0.2

  - **Router 2:**

    enable

    configure terminal

    interface g0/0

    ip address 10.0.0.2 255.255.255.252

    no shutdown

    exit

    ip route 192.168.1.0 255.255.255.0 10.0.0.1

Q - 4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Hint: Think about router-on-a-stick or Layer 3 switch concepts.

➢ **Test communication between VLANs:**

- Connect one PC to Sales and another to Support.
- Try to ping the Support PC from the Sales PC.
- **Result:** The ping fails because the two VLANs are separate networks.
- **Solution:** Use **router-on-a-stick** or a **Layer 3 switch** to enable communication between VLANs.